

# Ryan Tang

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## SUMMARY

Student with experience in Python, SQL, and statistical modeling through academic and independent projects. Strong foundations in data structures and algorithms and experience building projects in Python, C++, and JavaScript. Interested in applying data analysis and problem-solving skills in a data-focused internship.

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## EDUCATION

University of California San Diego

B.S., Data Science

9/2023 - present

- Current GPA: 3.9

## SKILLS AND COURSEWORK:

- **Technical Skills**
    - Languages: Python, C++, C, Java, SQL, JavaScript, HTML/CSS, R
    - Frameworks & Libraries: NumPy, pandas, scikit-learn, PyTorch, HuggingFace Transformers, D3.js, Dask
    - Tools & Platforms: Git, GitHub, UNIX (gdb, valgrind, make), AWS, PostgreSQL, SQLite, Jupyter
  - **Data & Machine Learning:**
    - Supervised learning (linear models, decision trees, random forests, boosting), NLP
    - Feature engineering, model evaluation (cross-validation, MSE, RMSE)
  - **Databases:**
    - Advanced SQL queries, joins, nested queries, aggregation, indexing
    - Relational algebra and database design principles
    - Exploratory data analysis and interactive visualizations using D3.js
    - Scalable analytics using cloud computing and parallel processing frameworks (AWS, Dask)
  - **Algorithms & Systems:**
    - Data structures, graph algorithms, dynamic programming, recursion, memory debugging
  - **Mathematical and Theoretical Foundations**
    - Probability & Statistics: Random variables, distributions, expectation/variance, CLT, parameter estimation (MLE, method of moments), hypothesis testing, confidence intervals
    - Stochastic Modeling: Markov chains, Hidden Markov Models, Bayesian networks, MDPs
    - Discrete Mathematics & Algorithms: Graph theory, combinatorics, algorithm analysis
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## PROJECTS:

### Interactive Clinical Data Visualization – D3.js, JavaScript

- Designed and implemented an interactive web visualization to explore perioperative clinical data
- Built dynamic filtering by age group and surgery type to compare mortality and ICU length of stay
- Processed and cleaned raw clinical datasets using Python (pandas)
- Developed responsive UI components with tooltips and scalable axes

### Video Game Review Rating Prediction – PyTorch, Transformers

- Designed and implemented an end-to-end NLP pipeline for large-scale review regression
- Built preprocessing, train/validation splits, and evaluation framework (MSE, RMSE)
- Integrated pretrained DistilBERT and implemented custom regression output layer
- Reduced error by ~57% relative to TF-IDF baseline